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(54) SPECTROSCOPY BASED ON SELECTIVE FREQUENCY BIN

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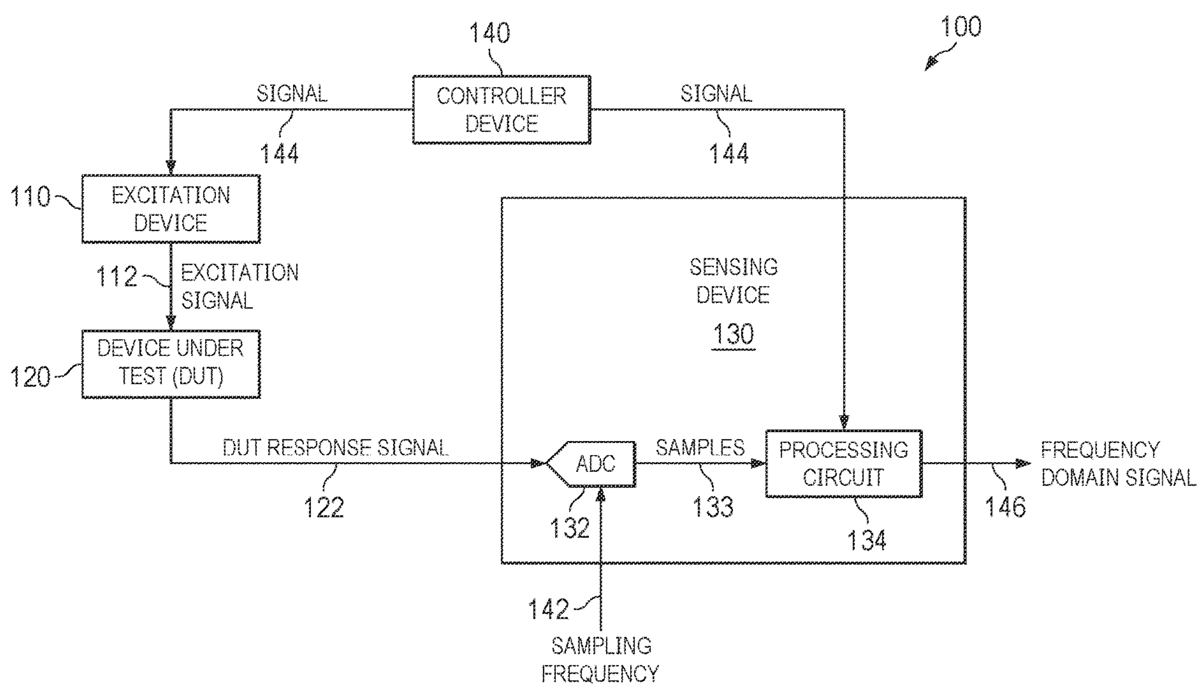
(57) **ABSTRACT**

An apparatus includes an analog-to-digital converter (ADC) circuit having an analog input, and a digital output, the ADC circuit configured to sample a first signal at the analog input at a sampling frequency and provide a second signal representing samples of the first signal at the digital output. The apparatus further includes a processing circuit having a processing input, a frequency bin width input, a frequency bin index input, and a processing output, the processing input coupled to the digital output, and the processing circuit configured to provide a third signal as a Fourier transform representation of at least a part of the samples at the processing output based on states of the frequency bin width input and the frequency bin index input.



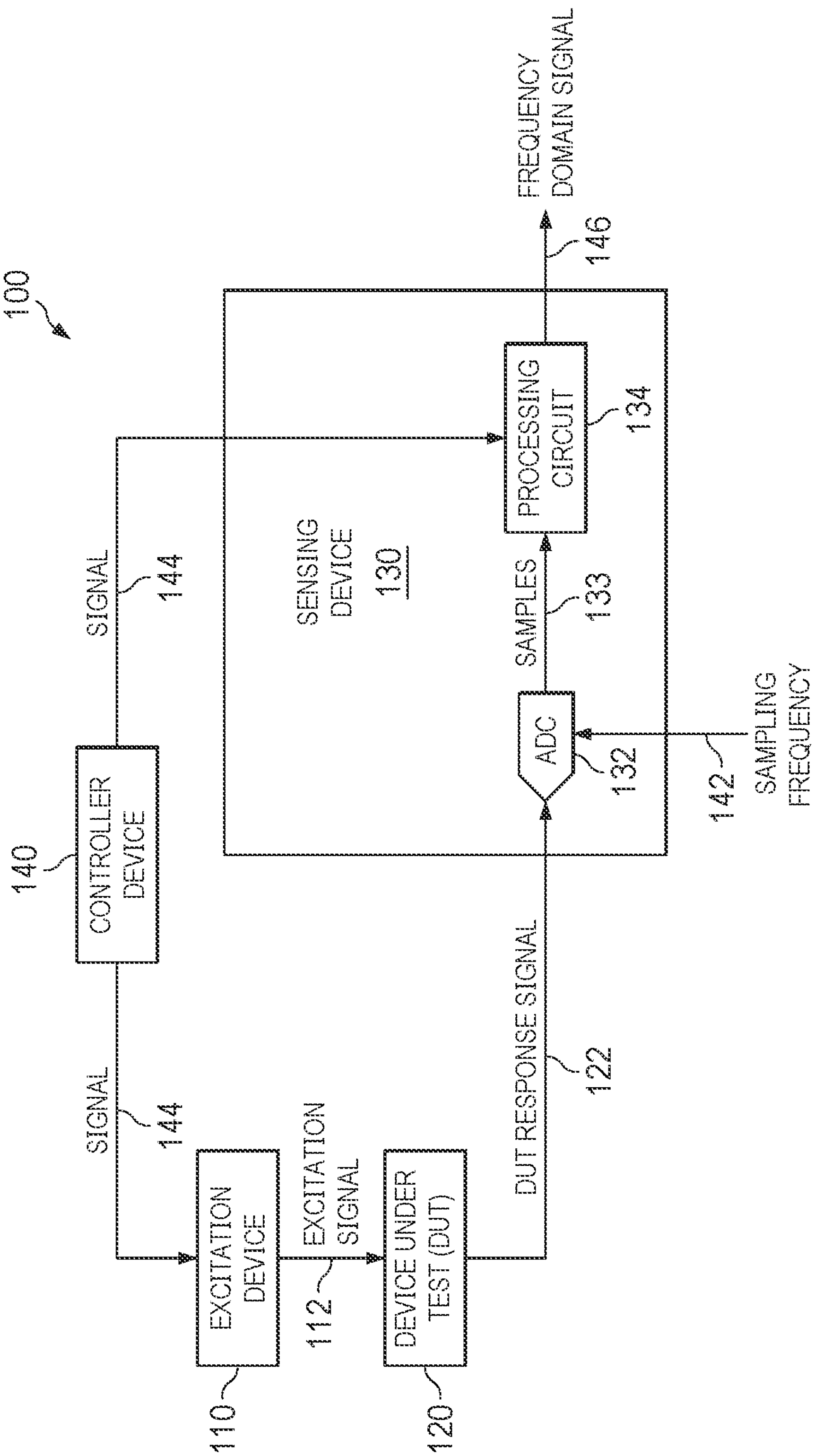

100
140
CONTROLLER DEVICE
SIGNAL
144
SIGNAL
144
110
EXCITATION DEVICE
EXCITATION SIGNAL
112
120
DEVICE UNDER TEST (DUT)
DUT RESPONSE SIGNAL
122
SENSING DEVICE
130
ADC
132
SAMPLES
133
PROCESSING CIRCUIT
134
142
SAMPLING FREQUENCY
146
FREQUENCY DOMAIN SIGNAL

FIG. 1

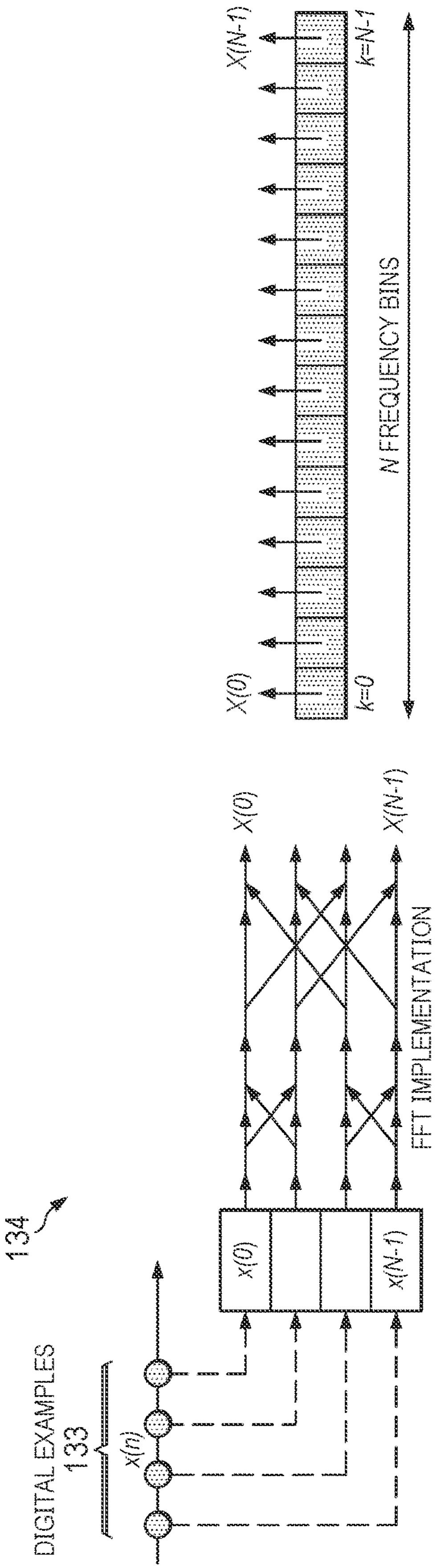
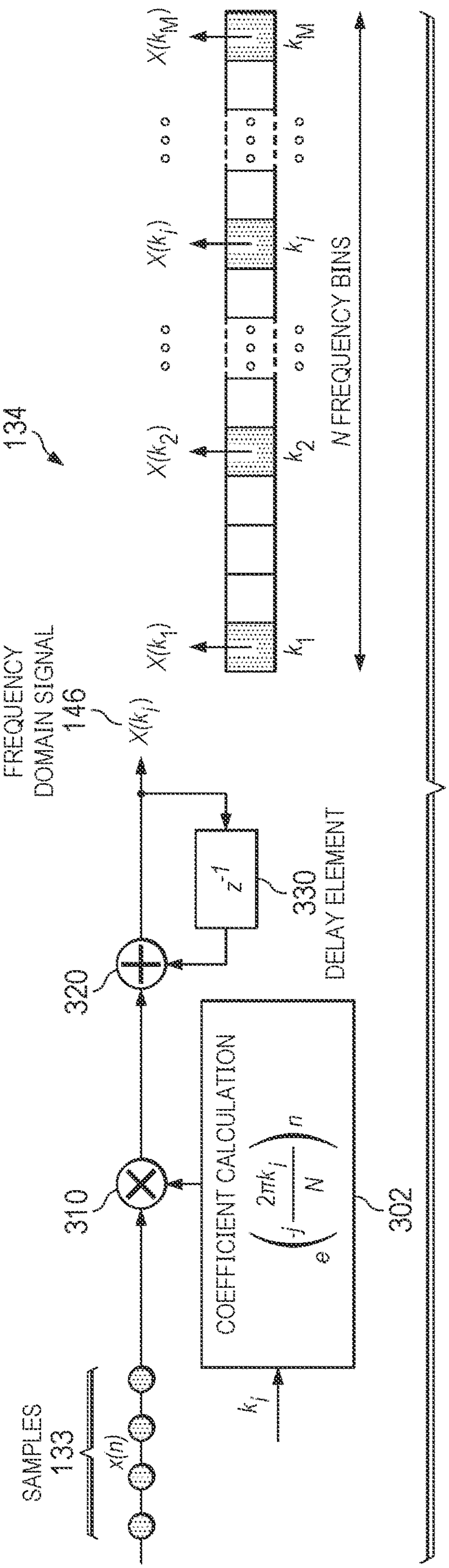
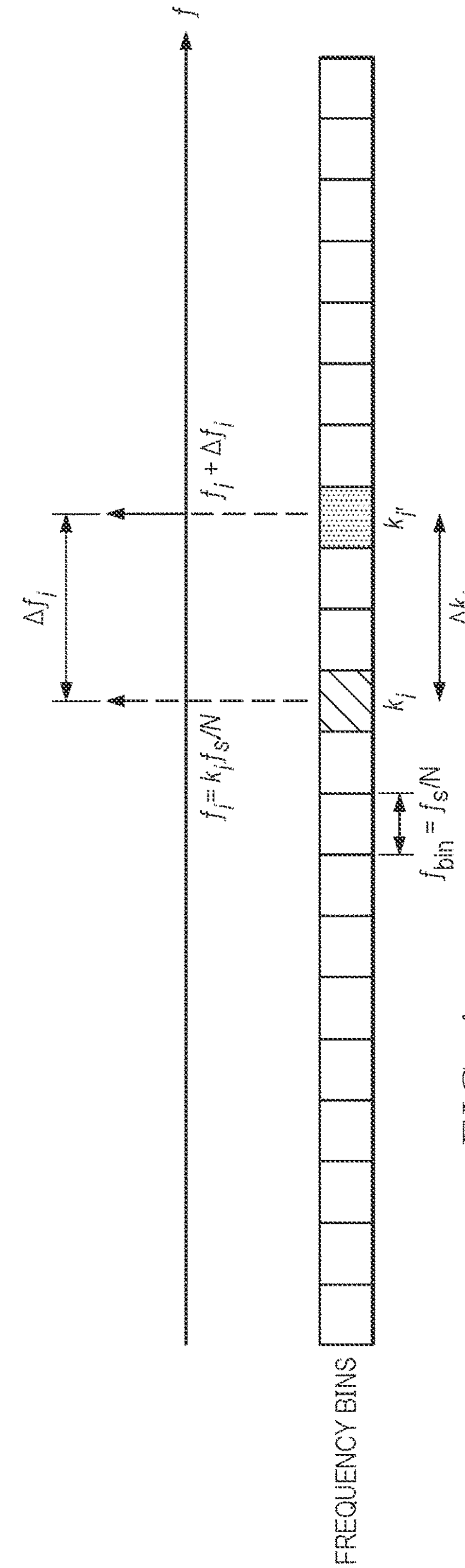

134
DIGITAL EXAMPLES
133
x(n)
x(0)
x(N-1)
FFT IMPLEMENTATION
X(0)
X(N-1)
X(0)
X(N-1)
k=0
k=N-1
N FREQUENCY BINS

FIG 2



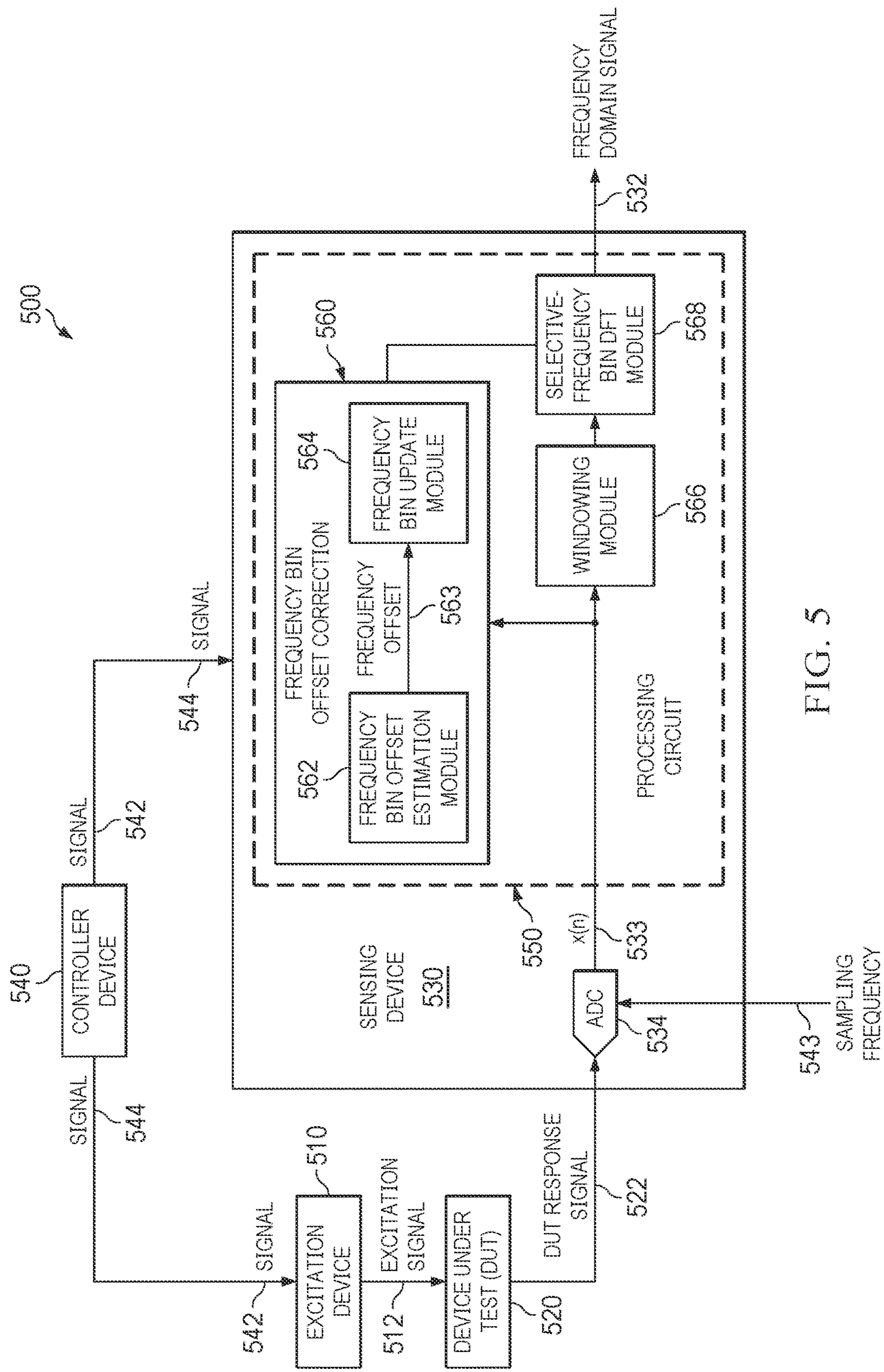
134
SAMPLES
133
x(n)
k_i
COEFFICIENT CALCULATION
$e^{\left(-j\frac{2\pi k_i}{N}\right)n}$
302
310
320
330
DELAY ELEMENT
z^{-1}
FREQUENCY DOMAIN SIGNAL
146
$X(k_i)$
$X(k_1)$
$X(k_2)$
$X(k_i)$
$X(k_M)$
k_1
k_2
k_i
k_M
N FREQUENCY BINS

FIG. 3



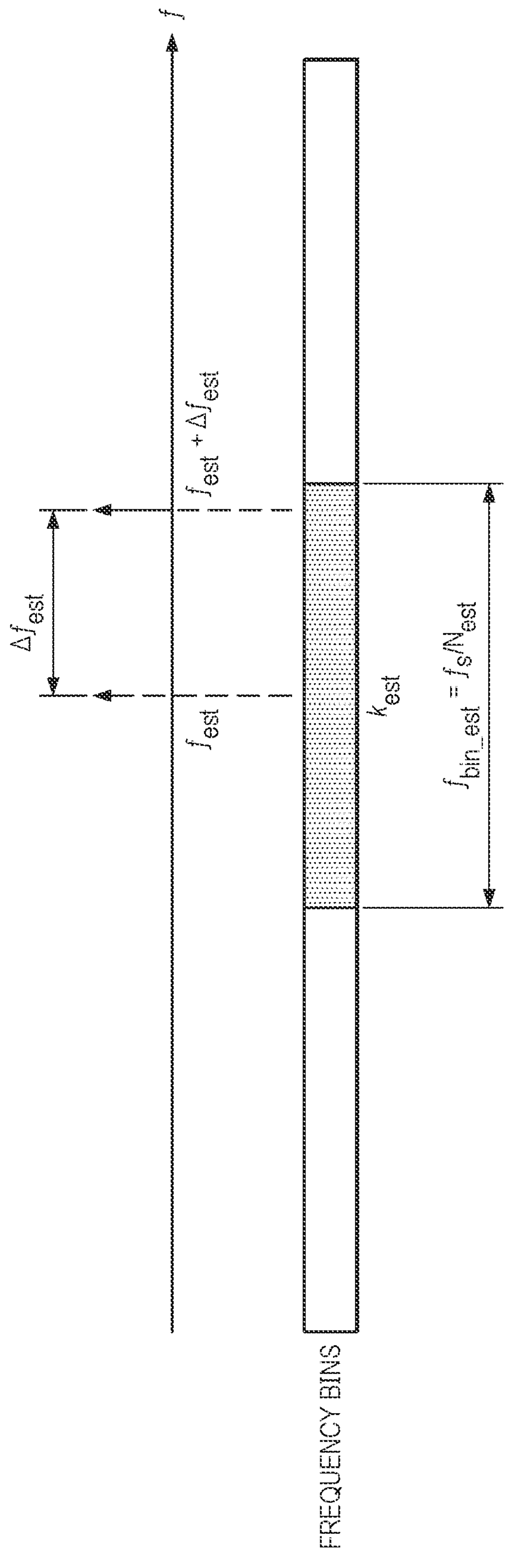
f
Δf_i
$f_i = k_i f_S/N$
$f_i + \Delta f_i$
k_i
k_i'
Δk_i
$f_{bin} = f_S/N$
FREQUENCY BINS

FIG. 4



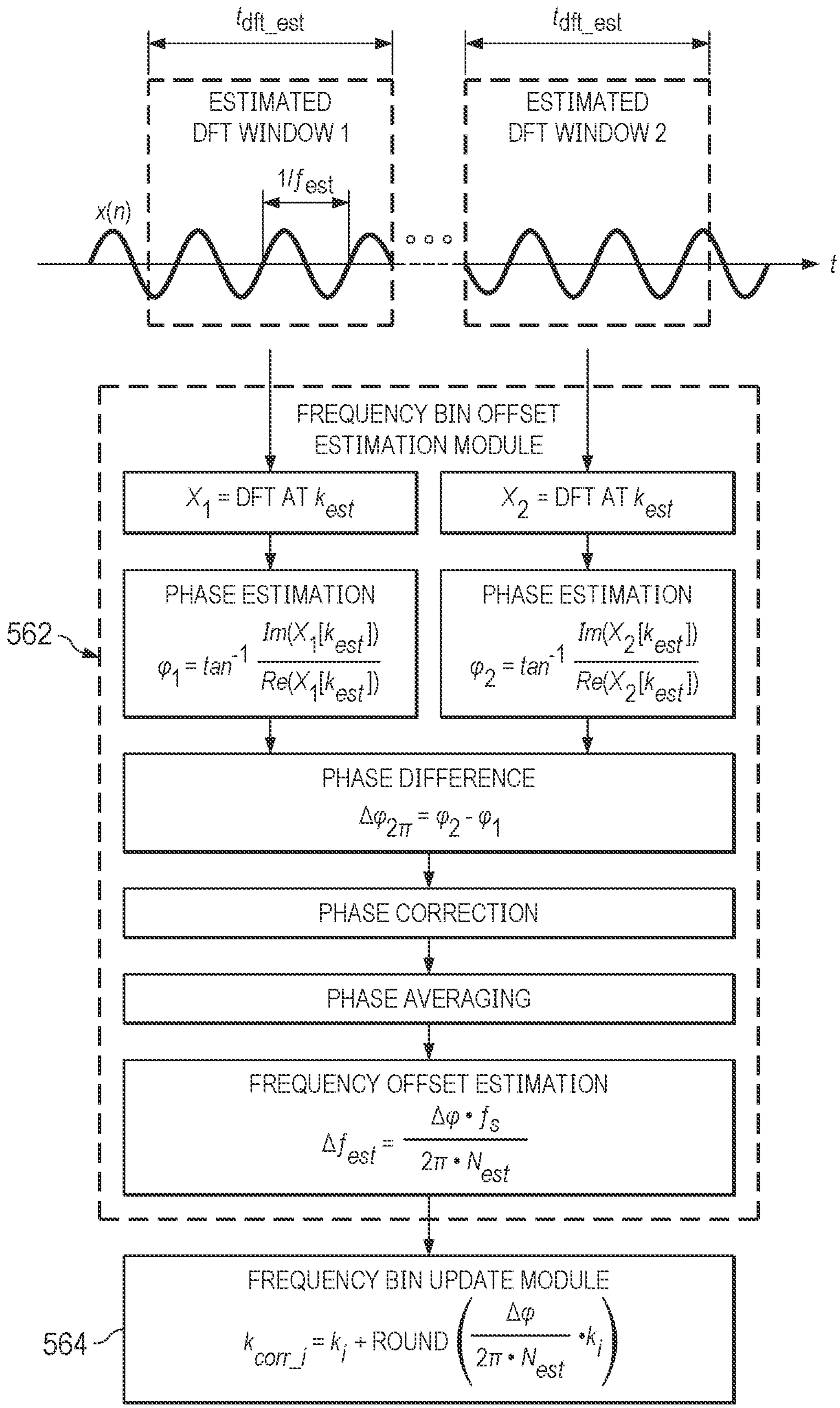
500
CONTROLLER DEVICE
540
SIGNAL
542
SIGNAL
544
SIGNAL
544
SIGNAL
542
EXCITATION DEVICE
510
EXCITATION SIGNAL
512
DEVICE UNDER TEST (DUT)
520
DUT RESPONSE SIGNAL
522
SENSING DEVICE
530
ADC
534
SAMPLING FREQUENCY
543
x(n)
533
550
PROCESSING CIRCUIT
FREQUENCY BIN OFFSET CORRECTION
560
FREQUENCY BIN OFFSET ESTIMATION MODULE
562
FREQUENCY OFFSET
563
FREQUENCY BIN UPDATE MODULE
564
WINDOWING MODULE
566
SELECTIVE-FREQUENCY BIN DFT MODULE
568
FREQUENCY DOMAIN SIGNAL
532

FIG. 5



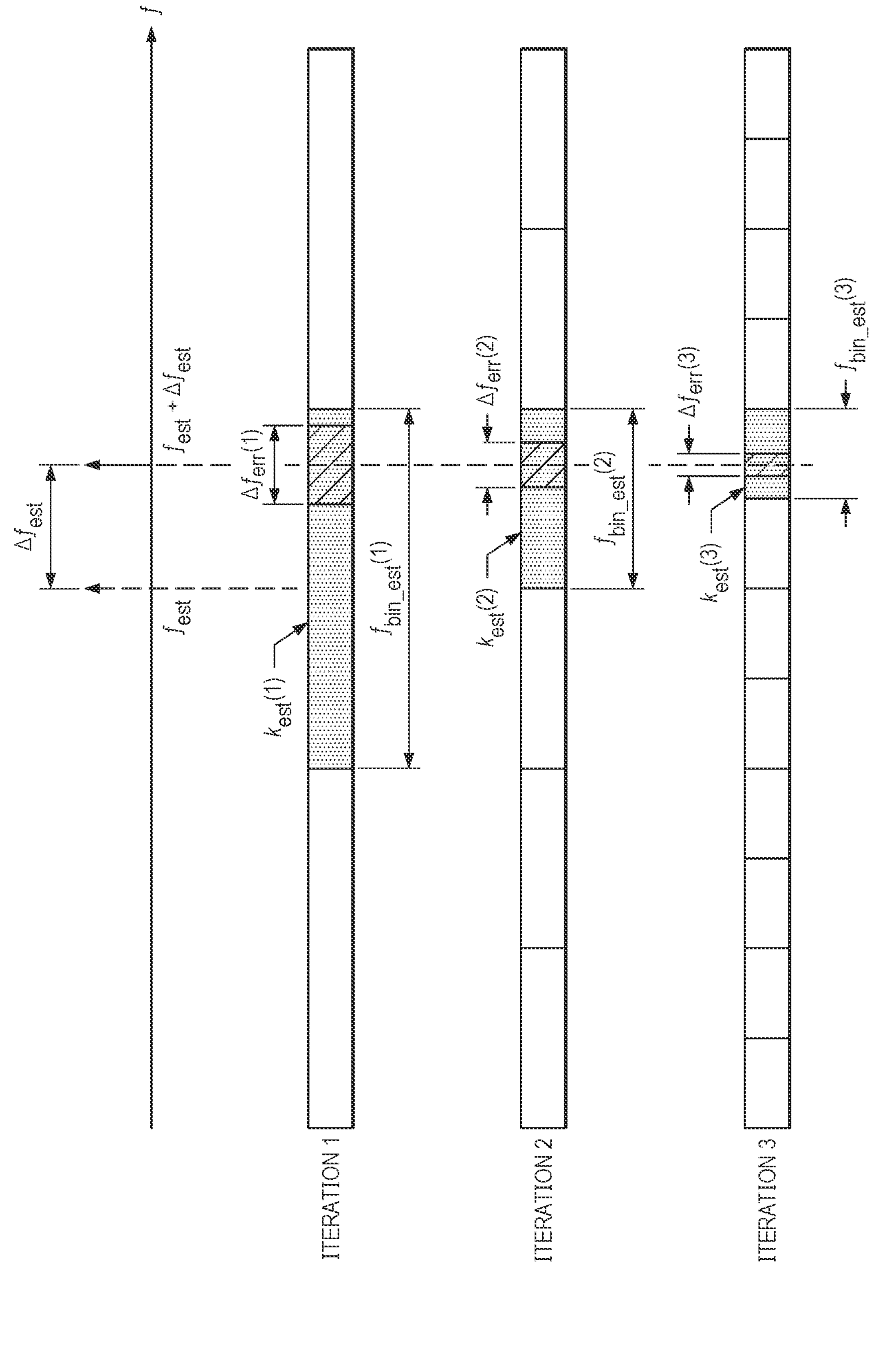
f
$f_{est} + \Delta f_{est}$
Δf_{est}
f_{est}
k_{est}
$f_{bin\_est} = f_s/N_{est}$
FREQUENCY BINS

FIG. 6



t_{dft_est}
t_{dft_est}
ESTIMATED DFT WINDOW 1
ESTIMATED DFT WINDOW 2
$1/f_{est}$
$x(n)$
t
FREQUENCY BIN OFFSET ESTIMATION MODULE
X_1 = DFT AT k_{est}
X_2 = DFT AT k_{est}
PHASE ESTIMATION
$\varphi_1 = tan^{-1} \frac{Im(X_1[k_{est}])}{Re(X_1[k_{est}])}$
PHASE ESTIMATION
$\varphi_2 = tan^{-1} \frac{Im(X_2[k_{est}])}{Re(X_2[k_{est}])}$
562
PHASE DIFFERENCE
$\Delta\varphi_{2\pi} = \varphi_2 - \varphi_1$
PHASE CORRECTION
PHASE AVERAGING
FREQUENCY OFFSET ESTIMATION
$\Delta f_{est} = \frac{\Delta\varphi \cdot f_s}{2\pi \cdot N_{est}}$
FREQUENCY BIN UPDATE MODULE
$k_{corr\_i} = k_i + \text{ROUND}\left(\frac{\Delta\varphi}{2\pi \cdot N_{est}} \cdot k_i\right)$
564

FIG. 7



f
Δf_{est}
f_{est}
$f_{est} + \Delta f_{est}$
$\Delta f_{err}(1)$
$k_{est}(1)$
$f_{bin\_est}(1)$
$\Delta f_{err}(2)$
$k_{est}(2)$
$f_{bin\_est}(2)$
$\Delta f_{err}(3)$
$k_{est}(3)$
$f_{bin\_est}(3)$
ITERATION 1
ITERATION 2
ITERATION 3

FIG. 8

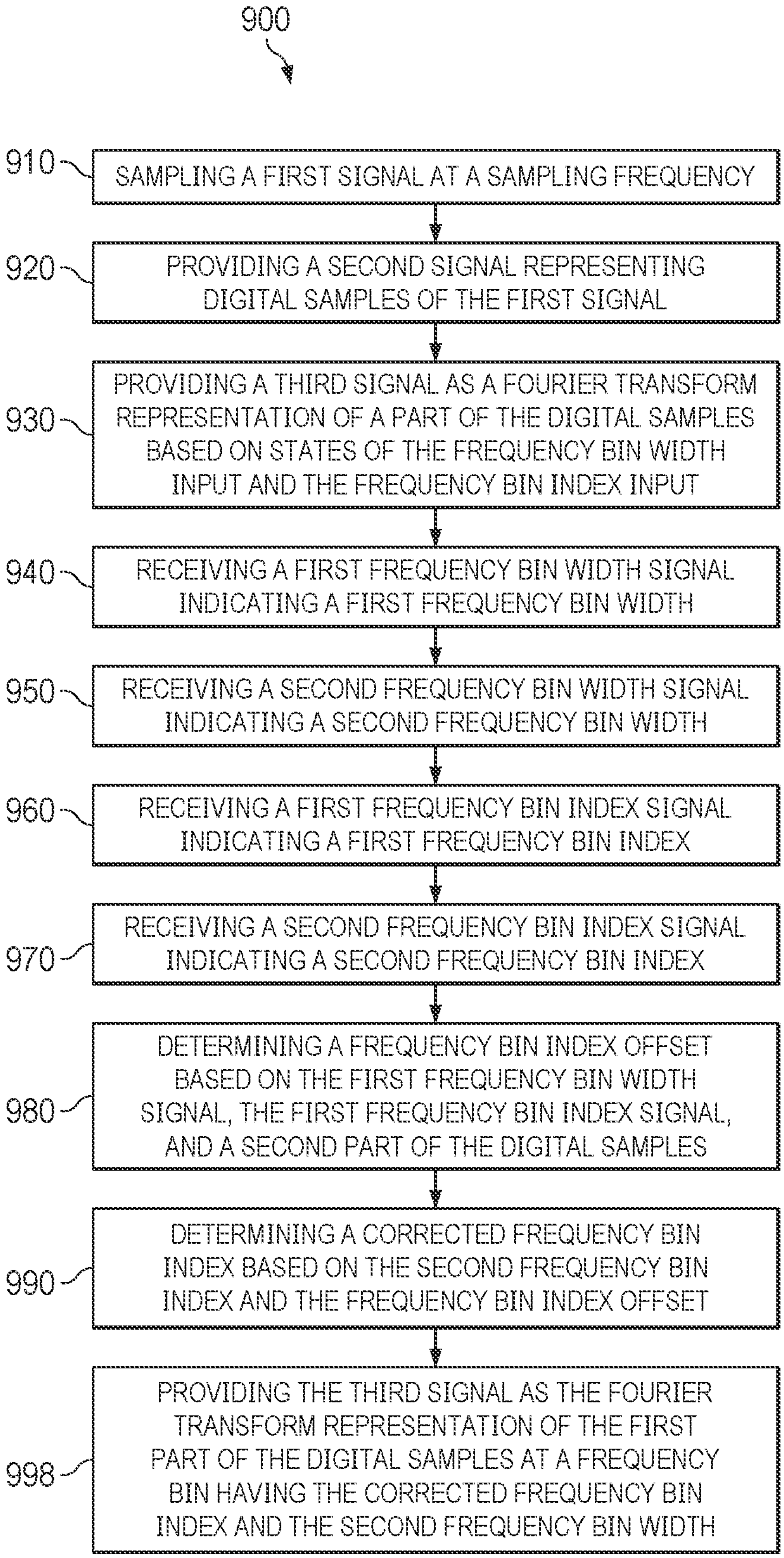

900
910
SAMPLING A FIRST SIGNAL AT A SAMPLING FREQUENCY
920
PROVIDING A SECOND SIGNAL REPRESENTING DIGITAL SAMPLES OF THE FIRST SIGNAL
930
PROVIDING A THIRD SIGNAL AS A FOURIER TRANSFORM REPRESENTATION OF A PART OF THE DIGITAL SAMPLES BASED ON STATES OF THE FREQUENCY BIN WIDTH INPUT AND THE FREQUENCY BIN INDEX INPUT
940
RECEIVING A FIRST FREQUENCY BIN WIDTH SIGNAL INDICATING A FIRST FREQUENCY BIN WIDTH
950
RECEIVING A SECOND FREQUENCY BIN WIDTH SIGNAL INDICATING A SECOND FREQUENCY BIN WIDTH
960
RECEIVING A FIRST FREQUENCY BIN INDEX SIGNAL INDICATING A FIRST FREQUENCY BIN INDEX
970
RECEIVING A SECOND FREQUENCY BIN INDEX SIGNAL INDICATING A SECOND FREQUENCY BIN INDEX
980
DETERMINING A FREQUENCY BIN INDEX OFFSET BASED ON THE FIRST FREQUENCY BIN WIDTH SIGNAL, THE FIRST FREQUENCY BIN INDEX SIGNAL, AND A SECOND PART OF THE DIGITAL SAMPLES
990
DETERMINING A CORRECTED FREQUENCY BIN INDEX BASED ON THE SECOND FREQUENCY BIN INDEX AND THE FREQUENCY BIN INDEX OFFSET
998
PROVIDING THE THIRD SIGNAL AS THE FOURIER TRANSFORM REPRESENTATION OF THE FIRST PART OF THE DIGITAL SAMPLES AT A FREQUENCY BIN HAVING THE CORRECTED FREQUENCY BIN INDEX AND THE SECOND FREQUENCY BIN WIDTH

FIG 9

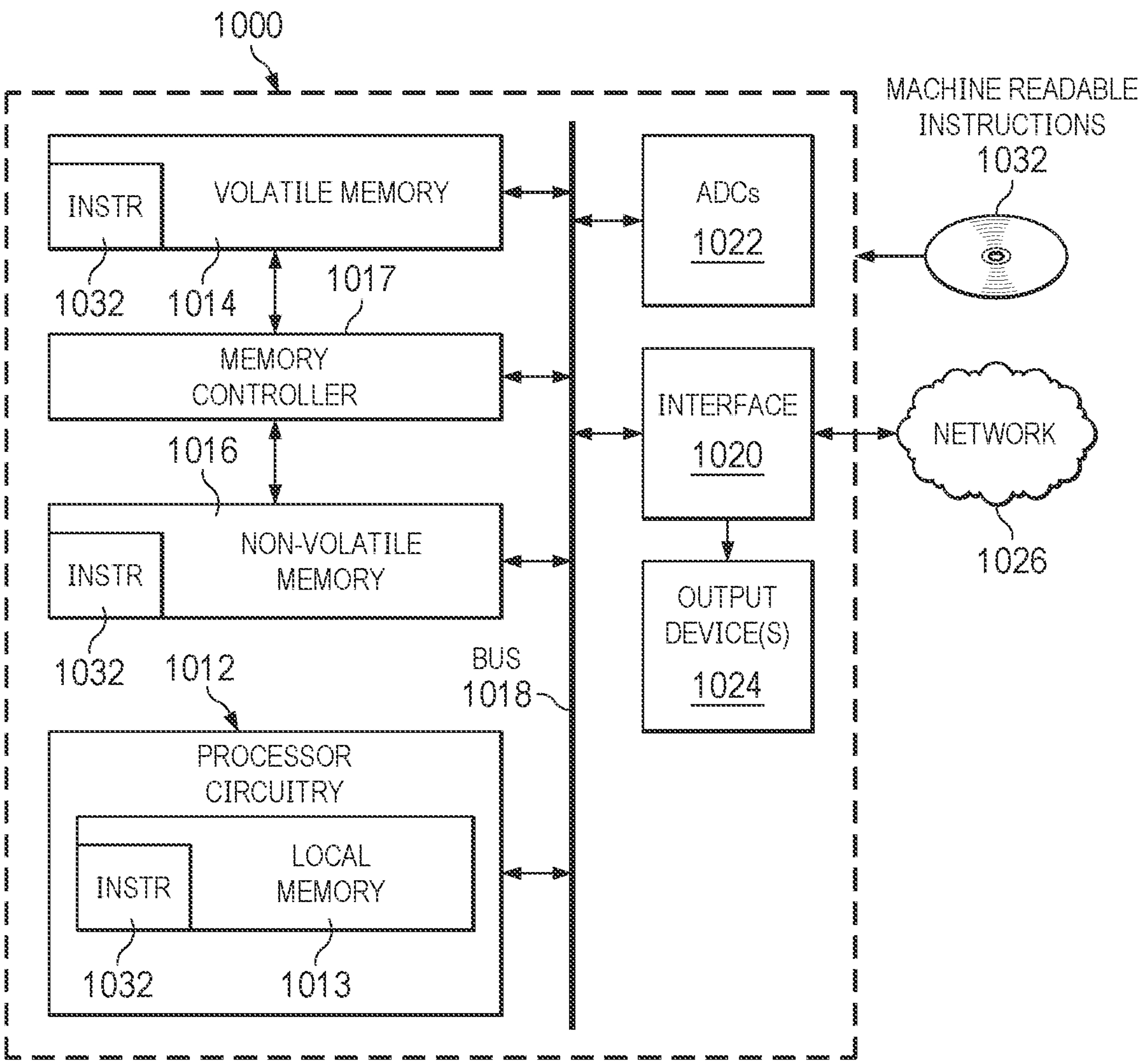

1000
MACHINE READABLE
INSTRUCTIONS
1032
INSTR
VOLATILE MEMORY
1032
1014
1017
ADCs
1022
MEMORY
CONTROLLER
INTERFACE
1020
NETWORK
1016
1026
INSTR
NON-VOLATILE
MEMORY
OUTPUT
DEVICE(S)
1024
1032
1012
BUS
1018
PROCESSOR
CIRCUITRY
INSTR
LOCAL
MEMORY
1032
1013

FIG. 10

SPECTROSCOPY BASED ON SELECTIVE FREQUENCY BIN

BACKGROUND

[0001] Spectroscopy is commonly used in several fields such as electrical spectroscopy, mechanical spectroscopy, optical spectroscopy and electrochemical impedance spectroscopy. One way of performing spectroscopy is based on analyzing the results of a Fourier transform, such as a discrete Fourier transform (DFT) of a signal, to analyze the properties of the signal in the frequency domain. For example, an excitation signal may cause a device under test to generate a response signal. The response signal can be sampled, and the samples can be converted from the time domain to the frequency domain using a DFT. An N-point DFT converts N number of time-domain samples into N number of DFT values, where each DFT value is a complex-valued, frequency domain representation of a signal at a particular frequency. A DFT may be implemented using a fast Fourier transform (FFT) where the FFT computes many DFT values in parallel. A FFT results in complex multiplications and additions as well as requiring a large amount of memory usage to store all of the complex DFT values. The complexity and memory usage involved in spectroscopy can be exacerbated in applications where a large number of DFT values are computed.

SUMMARY

[0002] In an example, an apparatus includes an analog-to-digital converter (ADC) circuit and a processing circuit. The ADC has an analog input, and a digital output. The ADC circuit is configured to sample a first signal at the analog input at a sampling frequency and provide a second signal representing samples of the first signal at the digital output. The processing circuit has a processing input, a frequency bin width input, a frequency bin index input, and a processing output. The processing input is coupled to the digital output, and the processing circuit is configured to provide a third signal as a Fourier transform representation of at least a part of the samples at the processing output based on states of the frequency bin width input and the frequency bin index input.

[0003] In an example, a system includes an excitation device having a first device under test (DUT) terminal. The system also includes a sensing device having a second DUT terminal. The system further includes a controller coupled to the excitation device and to the sensing device. The controller is configured to set a frequency of the excitation device. The controller is further configured to cause the excitation device to provide an excitation signal at the frequency at the first DUT terminal. The controller is additionally configured to transmit a frequency bin width signal, which can represent or can be based on the sampling frequency of the sensing device and the number of DFT values, and a frequency bin index signal to the sensing device, the frequency bin index signal being based on the frequency. The sensing device includes an analog-to-digital converter (ADC) circuit having an analog input and a digital output, the analog input coupled to the second DUT terminal. The ADC circuit is configured to sample a first signal at the analog input at a sampling frequency and provide a second signal representing samples of the first signal at the digital output. The sensing device further includes a processing circuit having a processing input, a frequency bin width input, a frequency bin index input, and a processing output, the processing input coupled to the digital output. The processing circuit is configured to provide a third signal as a Fourier transform representation of at least a part of the samples at the processing output based on the frequency bin width signal and the frequency bin index signal.

[0004] In an example, a method includes sampling a first signal at a sampling frequency. The method further includes providing a second signal representing samples of the first signal. In an example, the method also includes providing a third signal as a Fourier transform representation of a part of the samples based on states of a frequency bin width input and a frequency bin index input, wherein the at least the part of the samples is a first part of the samples. The method additionally includes receiving a first frequency bin width signal indicating a first frequency bin width. Additionally, the method includes receiving a second frequency bin width signal indicating a second frequency bin width. The method also includes receiving a first frequency bin index signal indicating a first frequency bin index. Furthermore, the method includes receiving a second frequency bin index signal indicating a second frequency bin index. The method additionally includes determining a frequency bin index offset based on the first frequency bin width signal, the first frequency bin index signal, and a second part of the samples. In an example, the method further includes determining a corrected frequency bin index based on the second frequency bin index and the frequency bin index offset. Moreover, the method further includes providing the third signal as the Fourier transform representation of the first part of the samples at a frequency bin having the corrected frequency bin index and the second frequency bin width.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic diagram of system for analyzing signal in frequency domain using a selective frequency bin, in an example.

[0006] FIG. **2** is a graph illustrating an FFT operation, in an example.

[0007] FIG. **3** is a schematic diagram of a DFT unit that performs a selective frequency bin DFT operation, in an example.

[0008] FIG. **4** is a graph illustrating a frequency bin offset, in an example.

[0009] FIG. **5** is a schematic diagram of a system for analyzing a signal in the frequency domain using a selective frequency with frequency bin offset correction, in an example.

[0010] FIG. **6** is a graph illustrating a frequency bin offset estimation operation, in an example.

[0011] FIG. **7** is a flowchart illustrating a frequency bin offset estimation and frequency update operation, in an example.

[0012] FIG. **8** is graph illustrating an iterative frequency bin offset estimation operation, in an example.

[0013] FIG. **9** is a flowchart of performing a spectroscopy analysis, in an example.

[0014] FIG. **10** is a schematic of an example processor that can implement part of the system of FIGS. **1**, **3**, **5**, and **7**-**9**, in an example.

DETAILED DESCRIPTION

[0015] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0016] FIG. **1** is a schematic diagram of system **100** for analyzing a signal in the frequency domain using a selective frequency bin DFT operation, in an example. The system **100** includes an excitation device **110** coupled to a device under test (DUT) **120** that is coupled to a sensing device **130**. The system **100** further includes a controller device **140** that is coupled to the excitation device **110** and to the sensing device **130**. The sensing device **130** includes an analog-to-digital converter (ADC) **132** that is coupled to a processing circuit **134**. Processing circuit **134** can include various modules, which can be circuit modules (e.g., part of an application specific integrated circuit (ASIC), a programmable logic array circuit), or software modules executable on a controller. The ADC **132** has an analog input and a digital output such that the analog signal, e.g., DUT response signal **122**, is received and converted into the discrete-time domain samples, e.g., samples **133**, and output from the digital output.

[0017] In an example, the DUT **120** is a target device that is analyzed in the frequency domain using a spectroscopy method, such as impedance spectroscopy. Examples of DUT **120** for which an impedance spectroscopy analysis is performed can include a charge storage device (e.g., a battery, a capacitor, a super capacitor, etc.), a motor, a sensor, etc. The excitation device **110** applies an excitation signal, e.g., electrical, mechanical, optical, etc., to the DUT **120** in order for the sensing device **130** to measure a DUT response signal **122** from the DUT **120**. The DUT response signal **122** may be converted to a voltage or current signal, as desired, by an analog front-end (not shown). The DUT response signal **112** is sampled at sampling frequency **142**, e.g., f_s, by the ADC **132** to convert the received analog signal into samples, e.g., samples **133**, that is subsequently processed by the processing circuit **134**. The processing circuit **134** of the sensing device **130** may include a DFT module that is implemented using a FFT, as shown in FIG. **2**. In an example, the frequency components of the signal calculated by the DFT module may be:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{\left(-j\frac{2\pi k}{N}\right)n} \quad \text{(Equation 1)}$$

[0018] In Equation 1, N is the number of DFT frequency bins and k is the frequency bin index from 0 to N−1. A frequency component is computed for each frequency bin associated with a particular frequency bin index k.

[0019] The number of complex multiplications for the DFT may be represented as $O(N^2)$, which illustrates that the complexity increases quadratically as N increases. The FFT computes many DFT components in parallel, as illustrated in FIG. **2**, for the samples **133**, which includes $(N/2)\log_2(N)$ complex multiplications and $N\log_2(N)$ complex additions. In certain applications, N is large. For example, in electrochemical impedance spectroscopy, the frequency may range from 0.025 Hz to 5 kHz to monitor all chemical reactions inside the battery. For a Nyquist ADC sampling rate of 10 kHz and a minimum frequency resolution of 0.025 Hz, **400,000** DFT values may be computed, which is process intensive as well as using a large amount of memory.

[0020] In some examples, to reduce the computation and the amount of memory usage, the controller device **140** can perform a selective frequency bin DFT operation, in which only the frequency components of a reduced number of frequency bins are computed. The frequency bins are selected based on their frequency bin index corresponding to a set of frequencies, such as the frequencies of the excitation signals. Such arrangements can reduce the number of frequency components computed, yet those frequency components can provide accurate representation of the response of the DUT to the excitation signals to support the spectroscopy analysis.

[0021] Specifically, to support the selective frequency bin DFT operation, the controller device **140** transmits a signal **144** to the excitation device **110** and the sensing device **130**. The signal **144** indicates the frequency bin width (e.g., a frequency width for each bin) and the frequency bin index. In other words, a particular frequency bin or bins are selected by the controller device **140**. Thus, the controller device **140** controls the excitation by the excitation device **110** and further controls the sensing device **130** by sending configuration parameters, e.g., frequency bin index, frequency bin width, etc., and control commands.

[0022] In response to receiving the signal **144**, the excitation device **110** generates an excitation signal **112** at the needed frequency components by the sensing device **130**. Accordingly, the excitation signal **112** generated by the excitation device **110** is aligned with the DFT module of the processing circuit **134** at the specified frequency bin, thereby reducing spectral leakage, increasing signal to noise ratio, and reducing memory usage. For example, the controller device **140** selecting a frequency bin results in computations for the selected bin instead of all frequencies, which occurs in a traditional FFT. In the electrochemical impedance spectroscopy, for the example provided above, 50 frequency components are used for the entire frequency range of 0.025 Hz to 5 kHz in order to estimate the battery parameters instead of using 400,000 frequency components. In an example, the excitation device **110** generates the excitation signal **112** at the frequencies of

$$f_i = \frac{k_i f_s}{N}$$

based on the frequency bin index and the frequency bin width f_{bin}, where

$$f_{bin} = \frac{f_s}{N},$$

i is an index for a particular frequency bin, f_s is the sampling frequency, and N is the number of frequency bins for the DFT module.

[0023] In an example, the processing circuit **134** includes a processing input that receives the samples **133**. The processing circuit **134** also includes a frequency bin width input and a frequency bin index input for receiving the signal **144**. The processing circuit **134** process the samples **133** using the configuration parameters received as signal **144** and provides a signal, e.g., frequency signal **146**, that is a Fourier transform representation of at least a part of the samples **133**, as shown by the equation below. The DFT is performed on a subset of frequencies instead of the entire frequency components. In an example, the DFT calculation by the processing circuit **134** based on the signal **144**, for a selective frequency bin DFT module, is shown below:

$$X[k_i] = \sum_{n=0}^{N-1} x[n] \cdot e^{\left(-j\frac{2\pi k_i}{N}\right)n} \quad \text{(Equation 2)}$$

[0024] The frequency domain signal **146** is output from the processing circuit **134** via its digital output. The frequency domain signal **146** represents properties of the DUT **120** at particular frequencies in the frequency domain.

[0025] FIG. **3** is a schematic diagram of a DFT module that performs a selective frequency bin DFT operation, in an example as shown. The processing circuit **134** may be implemented as a selective frequency bin DFT module as shown in FIG. **3**, for illustrative purposes that should not be construed as limiting the scope. The samples **133** are received, along with coefficients **302** and operated on by a multiplier **310**. The result is sent to an adder **320** where the result is summed with previous results via the delay element **330** until the frequency domain signal **146** is generated and output from the sensing device **130**. FIG. **3** is illustrated as one implementation of a selective frequency bin DFT module where each individual frequency component is computed serially using a multiplier **310** and an adder **320**, as illustrated. A coefficient for each frequency component may be generated by a coefficient generator at each sample instant n to multiply each sample **133**, e.g., x[n], by their corresponding coefficient. Accordingly, the DFT operation may be implemented efficiently for a small number of frequency components (e.g., since a subset of frequency bins rather than all frequency bins are computed) and a large number of DFT points. As such, the complexity is reduced by having only M number of multiplications and M−1 number of additions associated with the selected frequency bin, as opposed to the conventional DFT system. Moreover, the DFT computation is performed serially on the input time-domain samples, thereby eliminating the need for a large memory buffer.

[0026] As described above, the frequency components computed using a selective frequency bin DFT module can provide an accurate representation of the response of the DUT to the excitation signals to support the spectroscopy analysis if the selected frequency bins correspond to the frequencies of the excitation signals. But if there is a frequency bin offset between the selected frequency bins and the frequencies of the excitation signals, the output of the selective frequency bin DFT module may not accurately represent the response of the DUT.

[0027] FIG. **4** is a graph illustrating a frequency bin offset, in an example. Often times, separate devices are used as the excitation device **110** and the sensing device **130** with each having its own local clock. It is not uncommon for the local clock of different devices to drift with respect to one another, causing an offset. For example, in FIG. **4**, the controller device **140** may cause the excitation device **110** to generate an excitation signal **112** based on a selected frequency bin, e.g., k_i, having a particular frequency bin width with the excitation frequency at, e.g.

$$f_i = \frac{k_i f_s}{N}.$$

The controller device **140** communicates the same parameters to the sensing device **130**, e.g., signal **144**, and as such the sensing device **130** expects the signal at excitation frequency

$$f_i = \frac{k_i f_s}{N}.$$

However, the frequency may have drifted to $f_i+\Delta f_i$, where Δf_i is the frequency bin offset resulting from frequency drift between the sensing device **130** and the excitation device **110**. If the frequency bin offset is larger than the frequency bin width (e.g., $f_{bin}=f_s/N$), as illustrated in FIG. **4**, then the excitation signal **112** does not appear at the expected frequency bin k_i and instead may appear at a different frequency bin, e.g., k_i'. Accordingly, any measurement and/or computation by the sensing device **130** at the expected frequency bin index, e.g., k_i, results in a DFT value that is zero or close to because all of the signal energy is now at bin k_i' instead of k_i.

[0028] FIG. **5** is a schematic diagram of a system **500** for analyzing a signal in the frequency domain using a selective frequency with frequency bin offset correction, in an example. The system **500** can determine the amount of frequency shift and correct and update the frequency bin index to address the issue shown in FIG. **4**. The system **500** includes an excitation device **510** coupled to a DUT **520** that is coupled to a sensing device **530**. The system **500** further includes a controller device **540** that is coupled to the excitation device **510** and to the sensing device **530**. The sensing device **530** includes an ADC **534** that is coupled to a processing circuit **550**. Processing circuit **550** can include various modules, which can be circuit modules (e.g., part of an application specific integrated circuit (ASIC), a programmable logic array circuit), or software modules executable on a controller.

[0029] The ADC **534** has an analog input and a digital output such that analog signal, e.g., DUT response signal **522**, is received and converted into the discrete time domain, e.g., samples **133**, and output from the digital output. The processing circuit may include a frequency bin offset correction module **560** that is coupled to a windowing module **566** and a selective frequency bin DFT module **568** (all of which are described in greater detail below). The use of the selective frequency bin DFT module **568** is for illustration purposes and should not be construed as limiting the scope. For example, a multi-bin DFT module may be used where the number of bins is less than the total number of frequency components if a full DFT was performed.

[0030] In an example, the DUT **520** is a target device, e.g., a battery in electro-chemical impedance spectroscopy, human tissue in infrared spectroscopy, etc., that is analyzed in the frequency domain using a spectroscopy method. The excitation device **510** applies an excitation signal, e.g., electrical, mechanical, optical, etc., to the DUT **520** in order for the sensing device **530** to measure a DUT response signal **522** from the DUT **520**. The DUT response signal **522** may be converted to a voltage or current signal, as desired, by an analog front-end (not shown). The DUT response signal **512**

is sampled at a sampling frequency **543**, e.g., f_s, by the ADC **534** to convert the received analog signal to the discrete time domain samples, e.g., samples **533**, that are subsequently processed by the processing circuit **550**. The processing circuit **550** of the sensing device **530** may include a selective frequency bin DFT module **568**, as shown in FIG. **3**.

[0031] In order to reduce the number of frequency domain components (N), the number of computations and the amount of memory usage, the controller device **540** transmits a signal **542** to the sensing device **530**. The signal **542** indicates the frequency bin width (e.g., a frequency width of each bin) and the frequency bin index. In other words, a particular frequency bin or bins are selected by the controller device **540**. Thus, the controller device **540** controls the excitation by the excitation device **510** and further controls the sensing device **530** by sending configuration parameters, e.g., frequency bin index, and frequency bin width, which can represent or based on the sampling frequency of the sensing device and the number of DFT values, and control commands.

[0032] In response to receiving the signal **542**, the excitation device **510** generates an excitation signal **512** at one or more excitation frequencies

$$f_i = \frac{k_i f_s}{N},$$

where i is an index for a particular frequency bin, f_s is the sampling frequency, and N is the number of frequency bins for the DFT module. Excitation device **510** can compute the excitation frequency based on the frequency bin width f_s/N and frequency bin index k_i from signal **542**. Accordingly, the excitation signal **512** generated by the excitation device **510** is aligned with the DFT module of the processing circuit **550** at the specified frequency bin, thereby reducing spectral leakage, increasing signal to noise ratio, and reducing memory usage. For example, the controller device **540** selecting a frequency bin results in computations for the selected bin instead of all frequencies. In the electrochemical impedance spectroscopy, for the example provided above, 50 frequency components are used for the entire frequency range of 0.025 Hz to 5 kHz in order to estimate the battery parameters instead of using 400,000 frequency components in a full DFT implementation.

[0033] In an example, the processing circuit **550** includes a processing input that receives the samples **533**. The processing circuit **550** also includes a frequency bin width input and a frequency bin index input for receiving the signal **542**. The processing circuit **550** process the samples **533** using the configuration parameters received as signal **542** and forms a signal, e.g., a frequency domain signal **532**, that is a Fourier transform representation of at least a part of the samples **533**. In other words, the DFT is performed on a subset of frequencies instead of the entire frequency components. In an example, the DFT computation by the processing circuit **550** based on the signal **542**, for a selective frequency DFT module, is based on Equation 2 described above.

[0034] The frequency domain signal **532** is output from the processing circuit **550** via its digital output. The frequency domain signal **532** is indicative of analysis of properties of the DUT **520** in frequency domain.

[0035] As described above with respect to FIG. **4**, a frequency bin offset between the excitation device **510** and the sensing device **530** may cause the excitation signal **512** to not appear at the expected frequency bin k_i and instead may appear at a different frequency bin, e.g., k_i'. Accordingly, any measurement and/or computation by the sensing device **530** at the expected frequency bin, e.g., k_i, results in a DFT value that is zero or close to zero because all of the signal energy is now at bin k_i' instead of k_i. Accordingly, the controller device **540** may cause the sensing device **530** to perform a frequency bin offset calculation and a frequency update. The controller device **540** expands the frequency bin width to capture the frequency content that has moved outside of the expected frequency bin index due to frequency drift between the excitation device and the sensing device. For example, the controller device **540** may expand the frequency bin width to three times the original size such that the new bin width, k_i', would cover frequency bin index k_i, k_i+1, k_i+2. As such, the frequency content that could have been lost due to frequency bin offset can now captured in the new frequency bin for processing.

[0036] In order to expand the frequency bin width, the controller device **540** sends a signal **544** to the excitation device **510** and further to the sensing device **530**. The signal **544** indicates the offset estimation parameters, e.g., estimation frequency bin width (e.g., an expanded frequency width for each bin) and the estimation frequency bin index. The controller device **540** determines a particular frequency bin index, e.g., k_{est}, to be used by the excitation device **510** and the sensing device **530**. In an example, $f_{bin\_est}>2*\Delta f_{est}$ is the frequency bin width to be used during frequency bin offset estimation and where Δf_{est} is the largest frequency drift expected, so that that the frequency bin offset estimation frequency $f_{es}t$ can appear within the target frequency bin k_{est}. The controller device **540** may therefore determine N_{est}, which may be the number of DFT points based on the biggest expected frequency drift, and may determine

$$f_{bin\_est} = \frac{f_s}{N_{est}}$$

(in order to extend the frequency bin width). Accordingly, changing the number of DFT points changes the frequency bin width. As such, the frequency bin offset estimation frequency f_{est} is aligned with the frequency bin index k_{est} such that $f_{est}=k_{est}\cdot f_{bin\_est}$. The signal **544** transmits the frequency bin offset estimation parameters, e.g., estimation frequency bin index (k_{est}) and estimation frequency bin width (f_{bin_est}), or estimation frequency bin index (k_{est}) and the number of DFT values (N_{est}) to be used such that the excitation device **510** and/or the sensing device **530** can each compute the frequency bin width. Expansion of the frequency bin width for frequency bin offset estimation is shown in FIG. **6**. Increasing the frequency bin width results in the frequency bin offset to become a fraction of a known frequency bin width and can be estimated from the phase difference between two or more DFT measurements, which will be described below.

[0037] The excitation device **510** therefore receives the signal **544** that indicates the frequency bin offset estimation parameters, e.g., frequency bin index (k_{est}) and frequency bin width (f_{bin_est}), or estimation frequency bin index (k_{est}) and the number of DFT values (N_{est}) that can be used to compute the frequency bin width when the sampling frequency is known. The excitation device **510** generates excitation signal **512** similar to before and causes the DUT **520** to generate a DUT response signal **522** similar to before but with respect to the new excitation signal generated based on the offset estimation parameters. For example, the excitation device **510** generates and transmits the excitation signal **512** at f_{est} that corresponds to the frequency bin index k_{est} for a time duration that is at least $2N_{est}$ samples. The DUT response signal **522** may be converted to voltage or current, as desired, by an analog front-end (not shown) and subsequently converted to discrete time-domain by the ADC **534** using the sampling frequency **543**. The samples (x(n)) **533** are output from the digital output of the ADC **534** to the processing input of the processing circuit **550**, e.g., to the frequency bin offset correction module **560** and further to the windowing module **566**.

[0038] The frequency bin offset correction **560** of the processing circuit **550** within the sensing device **530** may include a frequency bin offset estimation module **562** and a frequency bin update module **564**. The frequency bin offset correction **560** may also receive the signal **544** as its input, e.g., frequency bin width input, frequency bin index input, etc. To determine the frequency bin offset estimation, the frequency bin offset estimation module **562** in an example, uses the time shift of continuous time Fourier Transform due to a time shift tO is shown below:

$$x(t-t_0)=e^{-(wt_0)}X(j\omega) \qquad \text{(Equation 3)}$$

[0039] The controller device **540** causes the processing circuit **550** to run two DFT windows, e.g., back-to-back, in order to estimate the frequency bin offset Δf_{est}. The time shift

$$t_0=\frac{N_{est}}{f_s}$$

for the time duration of one DFT window with N_{est} number of DFT points. The DFT windows size are chosen such that each window does not contain complete cycles of DUT response signal **122**, which allows the back-to-back DFT windows to capture DUT response signals having different starting phases. The frequency ω is calculated as $2\pi(f_{est}+\Delta f_{est})$ for a frequency bin offset estimation signal at frequency f_{est} with frequency bin offset Δf_{est}. As such, the phase shift between the two DFT windows may be represented as:

$$\Delta\varphi=\omega t_o=\frac{2\pi(f_{est}+\Delta f_{est})N_{est}}{f_s} \qquad \text{(Equation 4)}$$

[0040] In Equation 4, because $f_{est}=k_{est}f_{bin\_est}$ and

$$f_{bin\_est}=\frac{f_s}{N_{est}}$$

then

$$f_{est}=k_{est}\frac{f_s}{N_{est}}, \text{ and } \frac{2\pi f_{est}N_{est}}{f_s}=2\pi k_{est}.$$

Since k_{est} is an integer, the phase shift $\Delta\varphi$ can be calculated as follows:

$$\Delta\varphi=\frac{2\pi\Delta f_{est}N_{est}}{f_s} \qquad \text{(Equation 5)}$$

[0041] From Equations 4 and 5, the frequency bin offset Δf_{est} can be determined as follows:

$$\Delta f_{est}=\frac{\Delta\varphi f_s}{2\pi N_{est}} \qquad \text{(Equation 6)}$$

[0042] In other words, the phase shift between the two DFT windows is used by the frequency bin offset estimation module **562** to determine the frequency bin offset, as described above. The frequency bin offset **563** signal as determined by the frequency bin offset estimation module **562** is output to the frequency bin update module **564** where the corrected frequency bin index is generated, e.g., k_{corr_i}, for each measurement frequency bin k_i is calculated. The signal associated with the corrected frequency bin index is transmitted to the selective frequency bin DFT module **568** to compute the frequency domain signal **532** (Fourier transform representation of a part of the samples at frequency bin with the corrected frequency bin index and frequency bin width).

[0043] According to an example, in frequency bin offset estimation, the windowing module **566** receives the samples x(n) **533** and selects at least two parts of the samples **533**, e.g., a first and a second windows. The selected data associated with each window is sent to the selective frequency bin DFT module **568** to calculate a DFT output at frequency bin index k_{est} and with N_{est} number of DFT points. The selective frequency bin DFT module **568** generates one DFT output for each window, e.g., X1 and X2. The DFT output is sent to the frequency bin offset correction **560** where the phase for each DFT output is determined. For example, the phase, e.g., φ_1 and φ_2, for the DFT output X1 and X2 may be determined as:

$$\varphi_1=\tan^{-1}\frac{\mathrm{Im}\,(X_1[k_{est}])}{\mathrm{Re}\,(X_1[k_{est}])} \text{ and } \varphi_2=\tan^{-1}\frac{\mathrm{Im}\,(X_2[k_{est}])}{\mathrm{Re}\,(X_2[k_{est}])} \qquad \text{(Equation 7)}$$

[0044] The frequency bin offset estimation module **562** may determine the phase difference, e.g., $\Delta\varphi_{2\pi}=\varphi_2-\varphi_1$. The phase difference may be from -2π to 2π and is converted to a range from $-\pi$ to π according to the example algorithm below:

```
if Δφ_2π < − π
  Δφ_n = Δφ_2π + 2π
else if Δφ_2π > + π
  Δφ_n = Δφ_2π − 2π
```

-continued

$$\text{else}\quad \Delta\varphi_n = \Delta\varphi_{2\pi}$$

[0045] According to one example, the process as described above, is repeated and an average may be taken to improve the accuracy of the phase estimation. As described above, the frequency bin offset Δf_{est} is estimated by the frequency bin offset estimation module **562** based on the phase difference. For example, the frequency bin offset can be computed as follows:

$$\Delta f_{est} = \frac{\Delta\varphi f_s}{2\pi N_{est}} \quad \text{(Equation 8)}$$

[0046] The signal **563** associated with frequency bin offset Δf_{est} is sent to the frequency bin update module **564** in order to determine the corrected frequency bin index. The corrected frequency bin index is used to update the frequency in the sensing device **530** when performing the DFT operation. The frequency bin width during sensing is

$$f_{bin} = \frac{f_s}{N}.$$

The frequency bin offset for a frequency bin index k_i represented as a frequency bin offset, as follows:

$$\Delta k_i = \text{round}\left(\frac{\Delta f_i}{f_{bin}}\right) = \text{round}\left(\frac{\Delta\varphi k_i}{2\pi k_{est}}\right) \quad \text{(Equation 9)}$$

[0047] Moreover, because

$$\Delta f_i = \frac{\Delta f_{est}}{f_{est}} \cdot k_i \cdot \frac{f_s}{N},\ f_{bin} = \frac{f_s}{N} \text{ and } f_{est} = k_{est} \cdot \frac{f_s}{N_{est}},$$

the corrected frequency bin index k_{corr_i} is computed based on the phase difference $\Delta\varphi$ as follows:

$$k_{corr\_i} = k_i + \text{round}\left(\frac{\Delta\varphi \cdot k_i}{2\pi \cdot k_{est}}\right) \quad \text{(Equation 10)}$$

[0048] The corrected frequency bin index, as determined by the frequency bin update module **564** is sent to the selective frequency-bin DFT module **568** such that the corrected frequency bin index can be used by the selective frequency bin DFT module **568** to accurately perform a DFT to generate the frequency domain signal **532**. In other words, the controller device **540** may cause the excitation device **510** to generate the excitation signal **512**. The DUT **520** generates the DUT response signal **522** in response to the excitation signal **512**. The DUT response signal **522** is sampled by the ADC **534** and sent to the selective frequency bin DFT module **568** that uses the corrected frequency bin index to compute the DFT at the corrected frequency bin value, instead of computing the DFT at the frequency of the excitation signal.

[0049] In some examples, the frequency bin offset correction may be initiated or triggered by the controller device **540**. In an example, the frequency bin offset correction may be initiated when the output of the selective frequency bin DFT module **568** is zero or close to zero that indicates that there has been a frequency bin offset that may have resulted in the frequency component signals ending up at a different frequency bin index than the one that was expected. The controller device **540** sends control signals and configuration parameters to the excitation device **510** and the sensing device **530**. In frequency correction mode, the controller device **540** sends the frequency bin offset estimation parameters (k_{est}, N_{est}) to the excitation device **510** and the sensing device **530**. The control signals and the configuration parameters are used by the excitation device **510** to generate the estimation signal at frequency $f_{est}=k_{est} \cdot f_s/N_{est}$. The controller device also controls the sensing device **530** by causing it to initiate the frequency bin offset estimation algorithm. In one example, the sensing device **530** performs the estimation DFT for two, back-to-back time intervals and the phase estimation step to calculate the phase difference $\Delta\varphi$ and frequency bin offset Δf. The determined phase difference and frequency bin offset is used by the selective frequency bin DFT module **568** to correct the frequency bin index, e.g., k_{corr_i}. As such, in operation, when the controller device **540** sends the measurement parameters (k_i, N) to the excitation device **510** and the sensing device **530**. The excitation device **510** generates the excitation signal **512** and the sensing device **530** performs measurements of the DUT response signal **522** by performing DFT using the selective frequency bin DFT module **568** at the corrected frequency bin index k_{corr_i}. According to one example, additional measurements can be repeated using the same frequency bin offset estimation if the magnitude of the DFT output $|X[k_{corr_i}]$ is greater than a minimum threshold X_{min}. This condition indicates that the frequency bin offset variations are less than the measurement frequency bin and the estimated frequency bin offset is still valid for correct measurements. If the DFT output is less than the threshold, then a new frequency bin offset correction is required and the entire frequency bin offset correction flow is repeated.

[0050] Accordingly, the system **500** determines the frequency correction by estimating the frequency difference between the excitation frequency generated by the excitation device **510** and the measured frequency by the sensing device **530**. The estimated frequency difference is used to correct and update the frequency and the frequency bin to be used by the sensing device **530** when a selective frequency bin DFT module **568** is used by the processing circuit **550** to reduce the number of frequency components, to reduce the amount of memory, and to reduce the amount of processing.

[0051] FIG. **7** is a flowchart illustrating a frequency bin offset estimation and frequency bin update operation, in an example. FIG. **7** shows two windows, windows 1 and 2 where window 1 is a first part of digital sample **533** and window 2 is a second part of the digital sample **533**. Windows 1 and 2 may be selected by the windowing module **566** and the windows 1 and 2 may be two back-to-back windows. As described above with respect to FIG. **5**, X_1 and X_2 are computed, the phase for each DFT output is estimated and a phase difference is determined. The phase correction is determined and in an example the process may be repeated, and the phase averaging may be performed for more accurate results. The frequency bin offset may be estimated using the phase difference and the corrected frequency bin index is determined.

[0052] FIG. **8** is a graph illustrating an iterative frequency bin offset estimation operation, in an example. The accuracy of the frequency bin offset estimation depends on the noise in the sensed signals in addition to the estimation frequency bin width f_{bin_est} that is used in the DFT to estimate the signal phase. In other words, the narrower the frequency bin width, e.g., f_{bin_est}, the higher the accuracy of the frequency bin offset estimation. However, a narrower frequency bin width, f_{bin_est}, results in a smaller frequency bin offset range that can be detected. Accordingly, an iterative frequency bin offset estimation may be used to enable the system to start with a larger frequency bin width initially for frequency correction and to reduce the frequency bin width with each iteration until the frequency bin width reaches a desired value. In other words, a wide frequency bin width is initially used to increase the detectable frequency bin offset range but one that may result in a lower accuracy to detect the correct frequency bin index that corresponds to the desired impedance measurement frequency bin width. During each iteration, the frequency bin width is reduced, thereby improving the accuracy of the frequency bin offset estimate. Therefore, each of the following iterations will update the estimation DFT parameters k_{est} and N_{est} in order to decrease the estimation DFT bin width f_{bin_est} successively, which will decrease the frequency bin offset estimation error as shown. The accuracy of frequency bin offset estimation improves as the number of iterations increases. As illustrated, in iteration 1, the frequency bin width is larger in comparison to subsequent iterations to ensure that the frequency drift resulting in the frequency components being placed in a different frequency bin index to be captured. The estimation from iteration 1 may be used in iteration 2 and the frequency bin width is reduced, and the process is repeated until the last iteration where the frequency bin width and the corrected frequency drift is determined for use by the selective frequency bin DFT module **568**.

[0053] In an example, the iterative method starts at i=0 to initialize the estimation DFT parameters with an initial frequency bin k_{est_ini} and an initial number of estimation DFT points N_{est_ini} according to target f_{est} as shown before. The iteration index is incremented, e.g., i+1, and the frequency bin index $k_{est}(i+1)$ is computed based on the estimated frequency and the frequency bin width of the previous iteration, the previous $f_{bin\_est}(i)$ is divided by two to center the estimated frequency in the new frequency bin. The frequency bin index $k_{est}(i+1)$ is shown as:

$$k_{est}(i+1) = \text{round}\left(\frac{2\hat{f}_{est}(i)}{f_{bin\_est}(i)}\right) \quad \text{(Equation 11)}$$

[0054] According to an example, the number of DFT points estimation and frequency bin width estimation at ith iteration is computed to be:

$$N_{est}(i+1) = \text{round}\left(\frac{k_{est}(i+1)f_s}{\hat{f}_{est}(i)}\right) \quad \text{(Equation 12)}$$

-continued

$$f_{bin\_est}(i+1) = \frac{f_s}{N_{est}(i+1)} \quad \text{(Equation 13)}$$

[0055] The new frequency bin offset estimation at iteration i+1 may be computed based on the previously updated estimation DFT parameters. The new frequency bin offset estimation has less error in comparison to the one(s) computed in the previous iteration(s). The new frequency bin offset estimation is shown as:

$$\Delta f_{est}(i+1) = \frac{\Delta\varphi(i+1)\cdot f_s}{2\pi\cdot N_{est}(i+1)} \quad \text{(Equation 14)}$$

$$\hat{f}_{est}(i+1) = k_{est}(i+1)f_{bin\_est}(i+1) + \Delta f_{est}(i+1) \quad \text{(Equation 15)}$$

[0056] The iterations can be repeated until, for example, a target frequency bin offset estimation accuracy is achieved, and/or the number of iterations reach a desired value.

[0057] FIG. **9** is a flowchart of a method **900** for performing a frequency domain analysis of a signal using a selective frequency bin for a DFT module and performing frequency bin offset estimation and update, in an example. In operation **910**, a first signal is sampled at a sampling frequency, as described in FIGS. **1** and **5**. In operation **920**, a second signal representing the samples of the first signal is provided, as described in FIGS. **1** and **5**. In operation **930**, a Fourier transform representation of a part of the digital sample is provided based on the states of the frequency bin width and the frequency bin index, as input, as described in FIGS. **1** and **5**. In operation **940**, a first frequency bin width signal indicating a first frequency bin width is received, as described in FIGS. **1** and **5** through **8**. At step **950**, a second frequency bin width signal indicating a second frequency bin width is received, as described in FIGS. **1** and **5** through **8**. In operations **960** and **970**, a first and a second frequency bin indices signal indicating the first and the second frequency bin indices are received respectively, as described in FIGS. **1** and **5** through **8**. In operation **980**, a frequency bin index offset is determined based on the first frequency bin width signal, the first frequency bin index signal, and a second part of the samples, as described in FIGS. **5** through **8**. In operation **990**, a corrected frequency bin index is determined based on the second frequency bin index and the frequency bin index offset, as described in FIGS. **5** through **8**. In operation **998**, the third signal is provided as the Fourier transform representation of the first part of the samples at a frequency bin having the corrected frequency bin index and the second frequency bin width, as described in FIGS. **5** through **8**.

[0058] FIG. **10** is a block diagram of an example processor platform **1000** including processor circuitry structured to execute machine-readable instructions to implement the logic depicted in the examples of FIGS. **1** through **9**. Processor platform **1000** of the illustrated example can include processor circuitry **1012**. The processor circuitry **1012** of the illustrated example includes hardware. For example, processor circuitry **1012** can be implemented by one or more integrated circuits, logic circuits, FPGAs, microprocessors, Central Processing Units (CPUs), Graphical Processing Units (GPUs), Digital Signal Processors (DSPs), and/or microcontrollers from any desired family or manufacturer. Processor circuitry **1012** can be implemented by one or more semiconductor-based (e.g., silicon-based) devices. In this example, processor circuitry **1012** can implement the controllers **140** and processing circuit **134** of FIG. **1** and controller device **540**, frequency bin offset correction module **560**, frequency bin offset estimation module **562**, frequency bin update module **564**, windowing module **566**, and selective frequency-bin DFT module **568** of FIGS. **5** and **7**.

[0059] Processor circuitry **1012** of the illustrated example can include a local memory **1013** (e.g., a cache, registers, etc.). Processor circuitry **1012** of the illustrated example is in communication with a computer-readable storage device such as a main memory including a volatile memory **1014** and a non-volatile memory **1016** by a bus **1018**. The volatile memory **1014** can be implemented by, for example, Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory **10016** may be implemented by programmable read-only memory, flash memory and/or any other desired type of non-volatile memory device. Access to the main memory **1014**, **1016** of the illustrated example can be controlled by a memory controller **1017**.

[0060] The processor platform **1000** of the illustrated example also includes interface circuitry **1020**. The interface circuitry **1020** may be implemented by hardware in accordance with any type of interface standard, such as an Inter-Integrated Circuit (I2C) interface, a Serial Peripheral Interface (SPI), an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.

[0061] In the illustrated example, one or more input ADCs **1022** are connected to bus **1018**. The ADCs **1022** can convert analog signals to digital signals for processing by the processor circuitry **1012**. ADCs **1022** can represent ADCs **132** and **534**.

[0062] One or more output devices **1024** can be connected to the interface circuitry **1020** of the illustrated example. The output device(s) **1024** can include circuits such as driver circuits.

[0063] Machine-readable instructions **1032** can be stored in volatile memory **1014** and/or non-volatile memory **1016**. Upon execution by the processor circuitry **1012**, the machine-readable instructions **1032** cause the processor platform **1000** to perform any or all of the functionality described herein attributed to the systems **100** and **500** and method **900**.

[0064] In this description, the term "couple" may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0065] Also, in this description, the recitation "based on" means "based at least in part on." Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0066] A device that is "configured to" perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0067] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. An apparatus comprising:

an analog-to-digital converter (ADC) circuit having an input and an output, the ADC circuit configured to sample a first signal at the input at a sampling frequency and provide a second signal representing samples of the first signal at the output; and

a processing circuit having a processing input, a frequency bin width input, a frequency bin index input, and a processing output, the processing input coupled to the output of the ADC circuit, and the processing circuit configured to provide a third signal as a Fourier transform representation of at least a part of the samples at the processing output based on states of the frequency bin width input and the frequency bin index input.

2. The apparatus of claim **1**, wherein the at least the part of the samples is a first part of the samples, and the processing circuit is configured to:

receive a first frequency bin width signal indicating a first frequency bin width;

receive a second frequency bin width signal indicating a second frequency bin width;

receive a first frequency bin index signal indicating a first frequency bin index;

receive a second frequency bin index signal indicating a second frequency bin index;

determine a frequency bin index offset based on the first frequency bin width signal, the first frequency bin index signal, and a second part of the samples;

determine a corrected frequency bin index based on the second frequency bin index and the frequency bin index offset; and

provide the third signal as the Fourier transform representation of the first part of the samples at a frequency bin having the corrected frequency bin index and the second frequency bin width.

3. The apparatus of claim **2**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first number of the samples to be included in determining one or more second Fourier transform representations of the second part of the samples; and

wherein the second frequency bin width signal indicates a second number of the samples to be included in determining the first Fourier transform representation of the first part of the samples.

4. The apparatus of claim **2**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first sampling frequency at which the ADC samples the first signal to generate the second part of the samples; and

wherein the second frequency bin width signal indicates a second sampling frequency at which the ADC samples the first signal to generate the first part of the samples.

5. The apparatus of claim **2**, wherein the Fourier transform representation is a first Fourier transform representation, and the processing circuit is configured to:

determine a second Fourier transform representation of a first subset of the second part of the samples at a frequency bin having the first frequency bin index and the first frequency bin width;

determine a first phase of the second Fourier transform representation;

determine a third Fourier transform representation of a second subset of the second part of the samples at the frequency bin;

determine a second phase of the third Fourier transform representation;

determine a difference between the first and second phases; and

determine the frequency bin index offset based on the difference and the second frequency bin index.

6. The apparatus of claim **5**, wherein the first subset of second part of the samples is in a first time window, the second subset of the second part of the samples is in a second time window next to the first time window.

7. The apparatus of claim **5**, wherein the processing circuit is configured to iteratively determine the frequency bin index offset by progressively reducing the first frequency bin width and updating the first frequency bin index.

8. A system comprising:

an excitation device having a first device under test (DUT) terminal;

a sensing device having a second DUT terminal and comprising:

an analog-to-digital converter (ADC) circuit having an input and an output, the input of the ADC circuit coupled to the second DUT terminal, the ADC circuit configured to sample a first signal at the input of the ADC circuit at a sampling frequency and provide a second signal representing samples of the first signal at the output of the ADC circuit; and

a processing circuit having a processing input, a frequency bin width input, a frequency bin index input, and a processing output, the processing input coupled to the output of the ADC circuit, and the processing circuit configured to provide a third signal as a Fourier transform representation of at least a part of the samples at the processing output based on a frequency bin width signal at the frequency bin width input and a frequency bin index signal at the frequency bin width input; and

a controller coupled to the excitation device and to the sensing device, the controller configured to:

set a frequency of the excitation device;

cause the excitation device to provide an excitation signal at the frequency at the first DUT terminal; and

transmit the frequency bin width signal and the frequency bin index signal to the sensing device, the frequency bin index signal being based on the frequency.

9. The system of claim **8**, wherein the at least the part of the samples is a first part of the samples, and wherein the controller is configured to:

transmitting a fourth signal to the sensing device to initiate a frequency and a phase update on the sensing device based on the third signal;

transmit a first frequency bin width signal indicating a first frequency bin width to the sensing device;

transmit a first frequency bin index signal indicating a first frequency bin index to the sensing device;

transmit a second frequency bin width signal indicating a second frequency bin width; and

transmit a second frequency bin index signal indicating a second frequency bin index; and

wherein the processing circuit is configured to:

determine a frequency bin index offset based on the first frequency bin width signal, the first frequency bin index signal, and a second part of the samples;

determine a corrected frequency bin index based on the second frequency bin index and the frequency bin index offset; and

provide the third signal as the Fourier transform representation of the first part of the samples at a frequency bin having the corrected frequency bin index and the second frequency bin width.

10. The system of claim **9**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first number of the samples to be included in determining one or more second Fourier transform representations of the second part of the samples; and

wherein the second frequency bin width signal indicates a second number of the samples to be included in determining the first Fourier transform representation of the first part of the samples.

11. The system of claim **9**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first sampling frequency at which the ADC samples the first signal to generate the second part of the samples; and

wherein the second frequency bin width signal indicates a second sampling frequency at which the ADC samples the first signal to generate the first part of the samples.

12. The system of claim **9**, wherein the Fourier transform representation is a first Fourier transform representation, and the processing circuit is configured to:

determine a second Fourier transform representation of a first subset of the second part of the samples at a frequency bin having the first frequency bin index and the first frequency bin width;

determine a first phase of the second Fourier transform representation;

determine a third Fourier transform representation of a second subset of the second part of the samples at the frequency bin;

determine a second phase of the third Fourier transform representation; and determine a difference between the first and second phases;

determine the frequency bin index offset based on the difference and the second frequency bin index.

13. The system of claim **12**, wherein the first subset of second part of the samples is in a first time window, the second subset of the second part of the samples is in a second time window next to the first time window.

14. The system of claim **12**, wherein the processing circuit is configured to iteratively determine the frequency bin index offset by progressively reducing the first frequency bin width and updating the first frequency bin index.

15. A method comprising:

sampling a first signal at a sampling frequency;

providing a second signal representing samples of the first signal;

providing a third signal as a Fourier transform representation of a part of the samples based on states of a frequency bin width input and a frequency bin index input, wherein the at least the part of the samples is a first part of the samples;

receiving a first frequency bin width signal indicating a first frequency bin width;

receiving a second frequency bin width signal indicating a second frequency bin width;

receiving a first frequency bin index signal indicating a first frequency bin index;

receiving a second frequency bin index signal indicating a second frequency bin index;

determining a frequency bin index offset based on the first frequency bin width signal, the first frequency bin index signal, and a second part of the samples;

determining a corrected frequency bin index based on the second frequency bin index and the frequency bin index offset; and

providing the third signal as the Fourier transform representation of the first part of the samples at a frequency bin having the corrected frequency bin index and the second frequency bin width.

16. The method of claim **15**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first number of the samples to be included in determining one or more second Fourier transform representations of the second part of the samples; and

wherein the second frequency bin width signal indicates a second number of the samples to be included in determining the first Fourier transform representation of the first part of the samples.

17. The method of claim **15**, wherein the Fourier transform representation is a first Fourier transform representation;

wherein the first frequency bin width signal indicates a first sampling frequency at which an analog to digital (ADC) samples the first signal to generate the second part of the samples; and

wherein the second frequency bin width signal indicates a second sampling frequency at which the ADC samples the first signal to generate the first part of the samples.

18. The method of claim **15**, wherein the Fourier transform representation is a first Fourier transform representation, and wherein the method further comprises:

determining a second Fourier transform representation of a first subset of the second part of the samples at a frequency bin having the first frequency bin index and the first frequency bin width;

determining a first phase of the second Fourier transform representation;

determining a third Fourier transform representation of a second subset of the second part of the samples at the frequency bin;

determining a second phase of the third Fourier transform representation;

determining a difference between the first and second phases; and

determining the frequency bin index offset based on the difference and the second frequency bin index.

19. The method of claim **18**, wherein the first subset of second part of the samples is in a first time window, the second subset of the second part of the samples is in a second time window next to the first time window, and wherein the method further comprises: iteratively determining the frequency bin offset by progressively reducing the first frequency bin width and updating the first frequency bin index.

20. The method of claim **18** further comprising iteratively determining the frequency bin index offset by progressively reducing the first frequency bin width and updating the first frequency bin index.

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